JERMAINE LIM

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EDUCATION

Temasek Polytechnic 2025 – 2028 (expected)

Diploma in Applied Artificial Intelligence

- Relevant modules: Machine Learning and Data Processing, Deep Learning and Object Recognition, Cloud Technologies for AI, Mobile App Development, Robotic Process Automation, Data Structures and Algorithms

Hai Sing Catholic School, *GCE O-Levels* 2024

TECHNICAL SKILLS

Languages: Python, Dart, JavaScript, HTML, CSS, SQL (MySQL), Java, C++ (basic)

AI and ML: scikit-learn, pandas, LangGraph, LangChain, LLM APIs (Groq, Gemini, Amazon Bedrock), computer vision (AWS Rekognition), sentiment analysis (TextBlob)

Cloud: AWS (Lambda, API Gateway, S3, RDS, Cognito, SNS, CloudFront, IoT Core), Firebase (Auth, Firestore)

Apps and automation: Flutter, Riverpod, Next.js, Streamlit, Gradio, UiPath Studio and Orchestrator

Tools: Git and GitHub, KNIME, Power BI, Figma, Arduino, Raspberry Pi

PROJECTS

CafeHop | *Flutter, Firebase, Groq API, NETS QR* github.com/jnreii/cafehop

- Developed a cross-platform Flutter app for discovering and booking Singapore home cafes, with Firebase Auth (email, Google, phone) and real-time booking management in Cloud Firestore.
- Implemented AI recommendations with Llama 3.1 (Groq API) personalised to each user's activity, with safeguards against hallucinated results and a rating-based fallback.
- Integrated NETS QR payments for booking deposits, interactive maps with live location, and automated booking reminders.

Back2U | *AWS Lambda, Rekognition, Bedrock, RDS* github.com/jnreii/back2u

- Built a serverless lost-and-found web app on AWS (Lambda, API Gateway, RDS, Cognito, S3) with a weighted matching algorithm that scores reports on item type, colour, location and description, and ranks the top 5 candidates for staff verification.
- Applied computer vision with Amazon Rekognition, training a custom image classifier on 623 images, and built automatic blurring of faces and personal details in uploaded photos.

GAIA Research Agent | *Python, LangGraph, Gemini, Groq* github.com/jnreii/gaia-agent

- Developed an autonomous AI research agent with LangGraph that uses web search, document reading, code execution and vision tools to answer multi-step questions, passing the Hugging Face AI Agents Course benchmark.
- Improved reliability under API rate limits with progress checkpointing, automatic retries and error recovery.

HelpdeskBot | *UiPath, Python, Telegram API* github.com/jnreii/helpdesk-bot

- Automated IT helpdesk ticket triage end to end with UiPath and Python, classifying incoming emails by category and ITIL priority using keyword scoring and sentiment analysis.
- Deployed on UiPath Orchestrator with scheduled and API triggers, sending instant Telegram alerts for urgent tickets and generating daily Word reports for management.

Vehicle Insurance Cross-Sell Predictor | *scikit-learn, Streamlit* live app

- Trained and compared Decision Tree, Random Forest and Gradient Boosting models on 381,000+ customer records, using class weighting and hyperparameter tuning to raise recall on the minority interested class from 0.30 to over 0.90.
- Deployed the final model as a Streamlit app for single and batch predictions, achieving 91% recall on a held-out test set of 114,000+ customers while cutting the contact list by about 60%.

CERTIFICATIONS

Agentic AI Certified Foundations Associate, Oracle 2026

Certificate of Excellence, Hugging Face Agents Course 2026

Building with the Claude API, Claude Academy (Anthropic) 2026

Basic Proficiency in KNIME Analytics Platform, KNIME 2025

LEADERSHIP AND ACTIVITIES

Honorary Secretary, *Applied AI Student Interest Group (AAISIG)*, *Temasek Polytechnic* AY2026/27

- Coordinate with external partners for the Digital Kampung community outreach event, and develop interactive activities on AI literacy and scam awareness.
- Plan and facilitate club events, activities and merchandise launches, prepare committee proposals and minutes, and represent the school at open houses and industry showcases.

INNOVA Hackathon, *Silver Stamps Passport* 2026

- Interviewed seniors at an Active Ageing Centre and co-developed a stamp-based engagement programme and volunteer dashboard to encourage regular attendance, presenting the solution in the final pitch.

STEAMunity Hackathon, *SUTD* 2025

- Built an Arduino-based classroom climate system for students with autism that monitors indoor and outdoor temperature, alerts teachers when thresholds are exceeded and adjusts windows with servo motors; handled the circuit prototyping and wiring.

Member, *DMPC*, *Temasek Polytechnic* 2025

- Programmed lighting cues on a grandMA2 console and rigged lighting fixtures for live stage productions.

President, *Infocomm Club*, *Hai Sing Catholic School* 2023 – 2024

- Led the club and managed live sound, operating Yamaha mixers, microphones and speakers, including signal routing and EQ.

WORK EXPERIENCE

Retail Assistant (part-time, ad hoc), *Various retailers through MCI Recruitment and RecruitFirst* 2025 – Present

- Assisted customers and supported store operations at fashion and skincare retail outlets on short-term assignments, adapting quickly to new products and procedures.

Barista, *Tsujiri* Dec 2024 – Apr 2025

- Prepared beverages and desserts in a fast-paced café, handled orders and payments on the POS system, recommended retail products and maintained food hygiene standards.